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UNIT- 04 (MATHS- 04)

Qu-01 If $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$, $P(A \cup B) = \frac{5}{8}$ then

Find the Value of $P(A \cap B)$ [AKTU - 2022-23]

Sol^n

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= \frac{1}{4} + \frac{1}{2} - \frac{5}{8}$$

$$= \frac{2+4-5}{8} = \frac{6-5}{8} = \frac{1}{8}$$

$$\boxed{P(A \cap B) = \frac{1}{8}}$$ Ans

Qu-02 A random Variable X has the Following Probability distribution Values of

X:	0	1	2	3	4	5	6	7
P(X):	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

(i) Find k [AKTU - 2022-23]

(ii) $P(X \geq 6)$

$\Rightarrow \sum P(x) = 1$

(i) $0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k - 1 = 0$

$10k^2 + 10k - k - 1 = 0$

$10k(k+1) - 1(k+1) = 0$

$(k+1)(10k-1)$

$(k+1) = 0 \quad k = -1 \neq$

$10k - 1 = 0 \quad \boxed{k = \frac{1}{10}}$

$10k - 1 = 0$

$10k = 1$

$\boxed{k = \frac{1}{10}}$

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(ii) $P(X \geq 6)$

$= 1 - P(X < 6)$

$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$

$= 1 - [0 + K + 2K + 2K + 3K + K^2]$

$= 1 - [8K + K^2]$ $\quad \therefore K = \frac{1}{10}$

$= 1 - \left[\frac{8}{10} + \frac{1}{100}\right]$

$= \frac{100 - 81}{100}$ $\quad \boxed{P(X \geq 6) = \frac{19}{100}} = 0.19$

OR

$P(X \geq 6) = P(X=6) + P(X=7)$

$= 2K^2 + 7K^2 + K$ $\quad K = \frac{1}{10}$

$= 2 \times \frac{1}{100} + 7 \times \frac{1}{100} + \frac{1}{10}$

$= \frac{9}{100} + \frac{1}{10}$

$= \frac{9 + 10}{100}$

$\boxed{P(X \geq 6) = \frac{19}{100}} = 0.19$ Ans

Qu-03 A bag contain 10 white and 15 black balls. Two balls are drawn in succession without replacement. What is the probality that first white and second is black?

[AKTU - 2022-23]

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10 - white	
15 - Black	Total ball = 25

$P(W) = \frac{10}{25}$

$P(B) = \frac{15}{25}$

A :- First is white
B :- Second is black

$P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right)$

$= \frac{10}{25} \times \frac{15}{24}$

$\boxed{P(A \cap B) = \frac{1}{4}}$

Qu-04 If the Function $f(x)$ is defined by $F(x) = Ce^{-x}$ $0 \leq x \leq \infty$. Find the Value of C which change $f(x)$ to a Probability density function [AKTU-2021-22]

Sol^n $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx = 1$

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$= c\left[\frac{e^{-x}}{-1}\right]_0^{\infty} = 1$ $\qquad e^{-x} = \frac{e^{-x}}{-1}$

$\boxed{c = 1}$ $\qquad -c\,[e^{-\infty}]_0^{\infty}$

$-c\,[e^{-\infty} - e^{0}]$ $\qquad -c[-1] = \boxed{c = 1}$

Qu-05 Moment Generating Fun^n of Binomial Distribution ? **[AKTU - 2012, 2015, 2018]**

Sol^n MGF about origin

$M_x(t) = E(e^{tx}) = \sum_{r=0}^{n} e^{tx}\, p(x) = {}^nC_r\, p^r q^{n-r}$

$$M_x(t) = \sum_{r=0}^{n} e^{tx}\, {}^nC_r\, p^r q^{n-r}$$

$\therefore$ Where $(q+p)^n = {}^nC_r\, p^x q^{n-x}$ $\qquad e^{tr} \cdot p^r = (e^t p)^r$

$$M_x(t) = \sum_{r=0}^{n} {}^nC_r\, (e^t p)^r\, q^{n-r}$$

$$\boxed{M_x(t) = (q + pe^t)^n}$$

$\therefore$ N independent trials

$$\boxed{\text{Binomial Dist}^n = N(q+p)^n}$$

OR

Freq^n of r successive event $= N \cdot P(X = r)$

$= N \cdot {}^nC_r\, p^r q^{n-r}$ **Ans**

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Qu-06 Fit a binomial distⁿ to the Following Frequency data.

x :	0	1	2	3	4
f :	30	62	46	10	2

[AKTU - 2011, 2012, 2021-22]

Solⁿ

x	f	xf
0	30	0
1	62	62
2	46	92
3	10	30
4	2	8
	$\Sigma f = 150$	$\Sigma fx = 192$

$$\boxed{\text{Binomial dist}^n = N(q+p)^n}$$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{192}{150} = 1.28$$

$$\boxed{\mu = \bar{x} = 1.28}$$

For B D : $\mu = np$

$$1.28 = 4 \times p$$

$$p = \frac{1.28}{4}$$

$$\boxed{p = 0.32}$$

We know that $p + q = 1$

$$q = 1 - p$$

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$q = 1 - 0.32$

$\boxed{q = 0.68}$

$\boxed{N = \Sigma f = 150}$

$B.D = N(q+p)^n$

$\boxed{B.D = 150\,(0.68 + 0.32)^4}$ Ans

Qu-07 If the Probⁿ of hitting a target is 10% and 10 shots are fired independently. What is the Probᵐ that the target will be hit atleast once?
[AKTU - 2017, 2019]

Solⁿ $P(\text{hitting target}) = 10\% = \frac{10}{100} = \frac{1}{10} = p$

$P(\text{non-hitting}) = 1 - \frac{10}{100}$ $\quad p + q = 1$, $q = 1 - p$

$= \frac{10-1}{10} = \frac{9}{10} = q$

$P(X = r) = {}^nC_r\, p^r q^{n-r}$

Atleast once $P(r \geq 1) = 1 - P(r = 0)$

$= 1 - {}^{10}C_0\, p^0 q^{10-0}$

$= 1 - \left(\frac{9}{10}\right)^2$

$= 0.6513$ Ans

p = Probⁿ of Success
q = Failure Probᵐ

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Qu-08 Out of 800 Families with 4 children each how many Families would be expected to have?

Freq^n

(i) 2 boys and 2 girls

(ii) atleast one boy

(iii) No girl

(iv) atmost two girl

Assume Equal Prob^m For boys & girls

[AKTU - 2014, 2017, 2018, 2022]

Sol^n $P(Boy) = \frac{1}{2}$, $P(Girl) = \frac{1}{2}$, $N = 800$

$B.D = N(q+p)^n$

$\boxed{B.D = 800(q+p)^n}$

(i) Expected 2 boys and 2 girls

$P(X=2) = N \cdot {}^nC_r p^r q^{n-r}$ $n = 4$

$= 800 \times {}^4C_2 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2$

$= 800 \times \frac{\lfloor 4}{\lfloor 2 \, \lfloor 2} \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2$ $\lfloor 2 = 2!$

$= {}^4C_2 = \frac{\lfloor 4}{\lfloor 2 \, \lfloor 2}$

$= \frac{4 \times 3 \times \cancel{\lfloor 2}}{\lfloor 2 \times \cancel{\lfloor 2}}$

$\boxed{P(X=2) = 300}$

(ii) Atleast one boys)

$P(X \geq 1) = 800 [P(X=1) + P(X=2) + P(X=3) + P(X=4)]$

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$= 800 \left[{}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 + {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 \right]$

$= 800 \times \frac{1}{16} \left[{}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 \right]$

$= 800 \times \frac{1}{16} \left[4 + 6 + 4 + 1 \right]$

$\boxed{P(X \geq 1) = 750}$

${}^4C_1 = \frac{\lfloor 4}{\lfloor 1 \, \lfloor 3} = \frac{4 \times \cancel{\lfloor 3}}{\cancel{\lfloor 3}} = 4$

(iii) having no girls i.e having 4 boys.

$N(P(X=4)) = 800 \, (P(X=4))$

$= 800 \, {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 \Big]$

$= 800 \times \frac{1}{16}$

${}^4C_4 = \frac{\cancel{\lfloor 4}}{\cancel{\lfloor 4} \, \cancel{\lfloor 0}} = 1$

$\boxed{P(X = 4) = 50}$

(iv) having ~~atleast~~ ~~2~~ atmost 2 girl i.e atleast 2-boys

$N(P X \geq 2) = 800 \left[P(X=2) + P(X=3) + P(X=4) \right]$

$= 800 \left[{}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 + {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 \right]$

$\boxed{P(X \geq 2) = 550}$ Ans

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Qu-09 If a random variable X follow a Poisson distⁿ such that $P(X=2) = 9\,P(X=4) + 90\,(X=6)$ Find mean and variance of X?
[AKTU-2022-23, 2018]

Solⁿ Poisson distⁿ $= \dfrac{\lambda^r e^{-\lambda}}{\lfloor r}$

$P(X=2) = 9\,P(X=4) + 90\,P(X=6)$

$r=2$

$$\frac{e^{-\lambda}\lambda^2}{\lfloor 2} = 9\,\frac{e^{-\lambda}\lambda^4}{\lfloor 4} + 90\,\frac{e^{-\lambda}\lambda^6}{\lfloor 6}$$

$$\cancel{e^{-\lambda}}\lambda^2\left[\frac{1}{\lfloor 2}\right] = \left[\frac{9}{\lfloor 4}\lambda^2 + \frac{90}{\lfloor 6}\lambda^4\right]e^{-\lambda}\cancel{\lambda^2}$$

$$= \frac{1}{2} = \frac{\cancel{9}^{3}}{4\cdot\cancel{3}\cdot 2\cdot 1}\lambda^2 + \frac{\cancel{90}^{\cancel{30}\,\cancel{6}\,3}}{\cancel{6}_{2}\cdot\cancel{5}\cdot 4\cdot\cancel{3}\cdot\cancel{2}\cdot 1}\lambda^4$$

$$= \frac{1}{2} = \frac{3\lambda^2}{8} + \frac{\lambda^4}{8} \qquad = \frac{1}{2} = \frac{3\lambda^2+\lambda^4}{8}$$

$$\frac{\cancel{8}^{4}}{\cancel{2}} = 3\lambda^2 + \lambda^4$$

$$= 4 = 3\lambda^2 + \lambda^4$$

$\lambda^4 + 3\lambda^2 - 4 = 0$

$\lambda^2 + 4\lambda^2 - \lambda^2 - 4 = 0$

$\lambda^2(\lambda^2+4) - 1(\lambda^2+4) = 0$

$\lambda^2 - 1 = 0$ $\qquad$ $\lambda^2+4=0$ [Imiginary Root]

$(\lambda+1)(\lambda-1)$

$\lambda^2+4=0$

$\lambda^2 = \sqrt{-4}$

$\boxed{\lambda = \pm 1}$ Ans = Mean & Variance

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Qu-10 In a Poisson dist^n $P(r)$ For $r=0$ is 10%
Find mean ?

Sol^n $P(r) = \frac{e^{-\lambda} \lambda^r}{\lfloor r}$ — (1)

$P(0) = 10\% = \frac{10}{100} = \frac{1}{10}$

$P(0) = \frac{e^{-\lambda} \cdot \lambda^0}{\lfloor 0}$

$e^{-\lambda} = \frac{1}{10}$

$e^{-\lambda} = \frac{1}{10}$

$10 = \frac{1}{e^{-\lambda}}$

$\boxed{10 = e^{\lambda}}$

$e^{\lambda} = 10$

$\boxed{\lambda = \log_e 10}$ Ans =

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MIND - MAP

Probability

Representation :- $P(A)$

Formula :- $P(A) = \frac{n(A)}{n(S)}$ → Favorable Cases
→ Possible Cases (Total)

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

Mean $= \mu = \frac{\Sigma p_i x_i}{\Sigma p_i}$

Variance $= \sigma^2 = \Sigma p (x_i - \mu)^2 = \mu_2$

Probability density function $= \int_{-\infty}^{\infty} f(x)\,dx = 1$

Binomial Distribution $P(X = r) = {}^nC_r p^r q^{n-r}$

Mean → $\mu = np$
Variance → $\mu_2 = npq$
S.D → $\sqrt{npq}$

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Poisson Distⁿ :- $P(X=r) = \frac{\lambda^r e^{-\lambda}}{\lfloor r}$

↓

Recurrence Relⁿ :- $P(r+1) = \frac{\lambda}{r+1} P(r)$

↓

Normal Distⁿ : $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

↓

Standard Normal Distⁿ is :- $f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}z^2}$

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